



› QUANTUM MACHINE LEARNING

Classification of Handwritten Digits

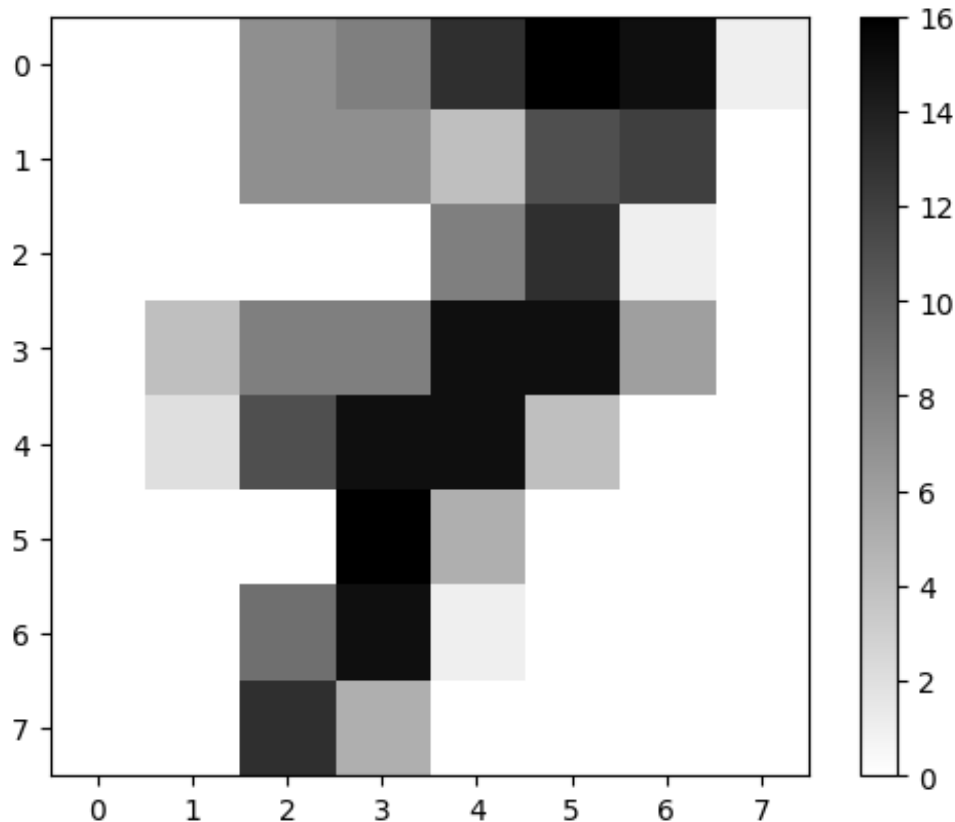
AGENDA

- > Dataset Exploration
- > Feature Reduction
- > Encoding Circuit Selection
- > Observable Selection
- > Final Pipeline & Evaluation
- > (Optional) Accelerated Hyperparameter Search

› DATASET EXPLORATION

Getting to know the data.

DATASET EXPLORATION



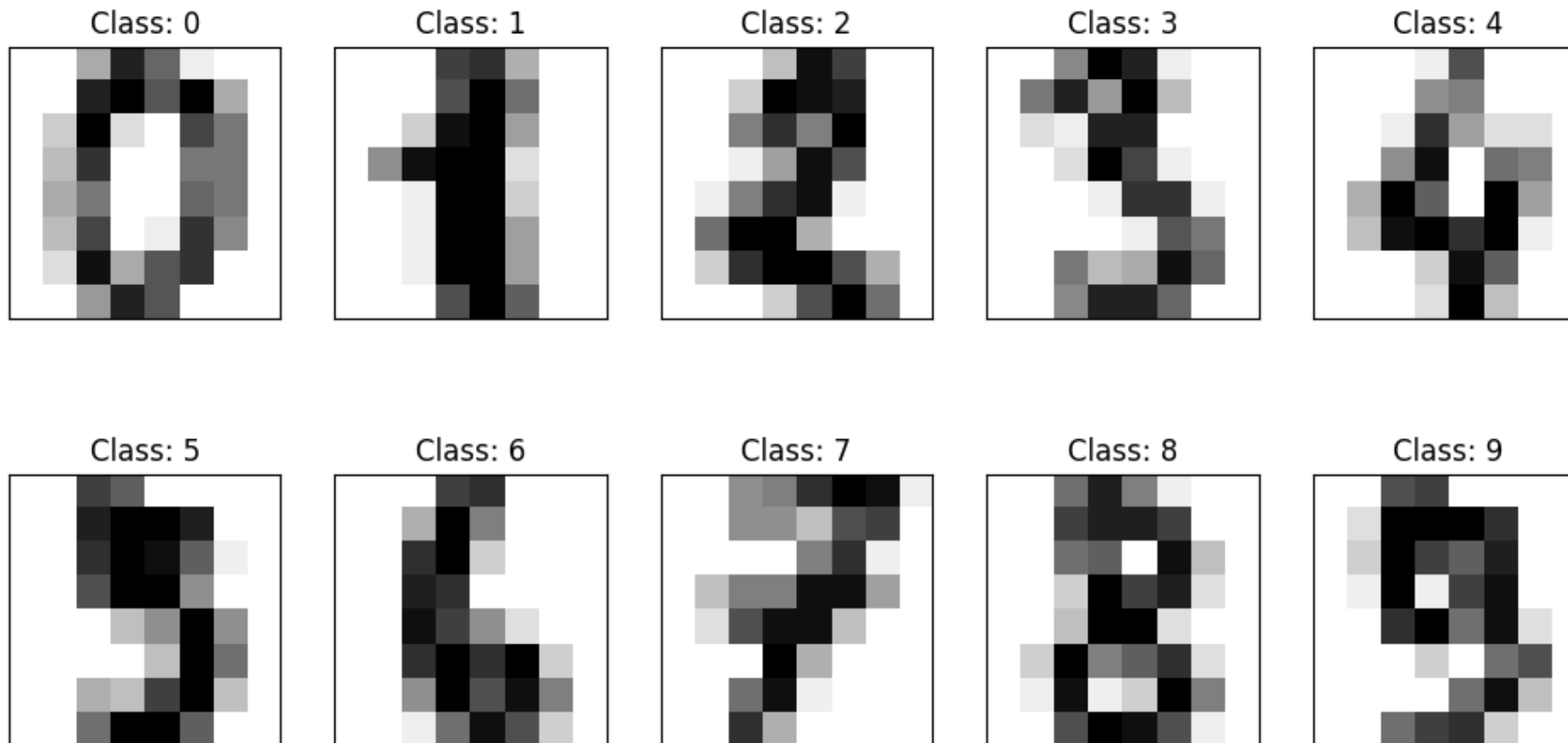
Each sample is represented as a 64-dimensional vector of pixel intensities, ranging from 0 to 16.

$$\vec{x} = [x_1 \ x_2 \ \dots \ x_{64}], \quad x_i \in [0; 16]$$

To reconstruct the image I from $\vec{x}$, we need to reshape the vector to an 8×8 matrix:

$$I = \begin{bmatrix} x_1 & \cdots & x_8 \\ \vdots & \ddots & \vdots \\ x_{57} & \cdots & x_{64} \end{bmatrix}$$

DATASET EXPLORATION

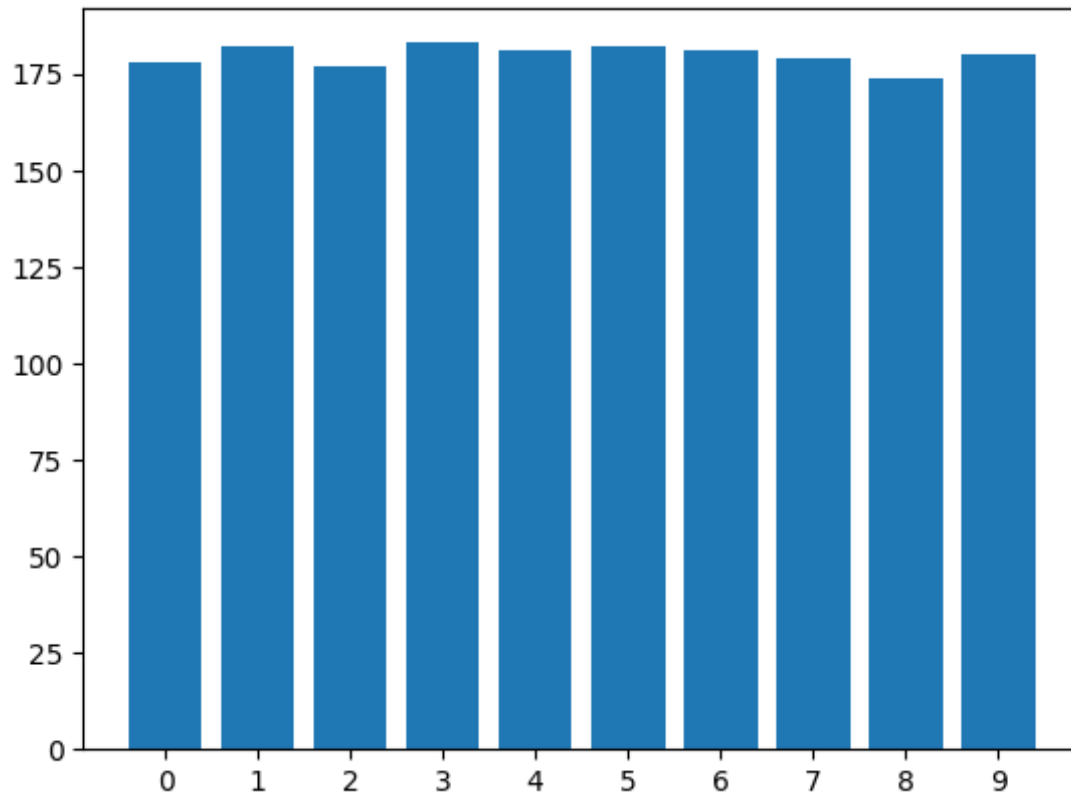


Each image is associated with a label

$$y \in \{0, 1, \dots, 9\}$$

representing the handwritten digit it contains.

DATASET EXPLORATION



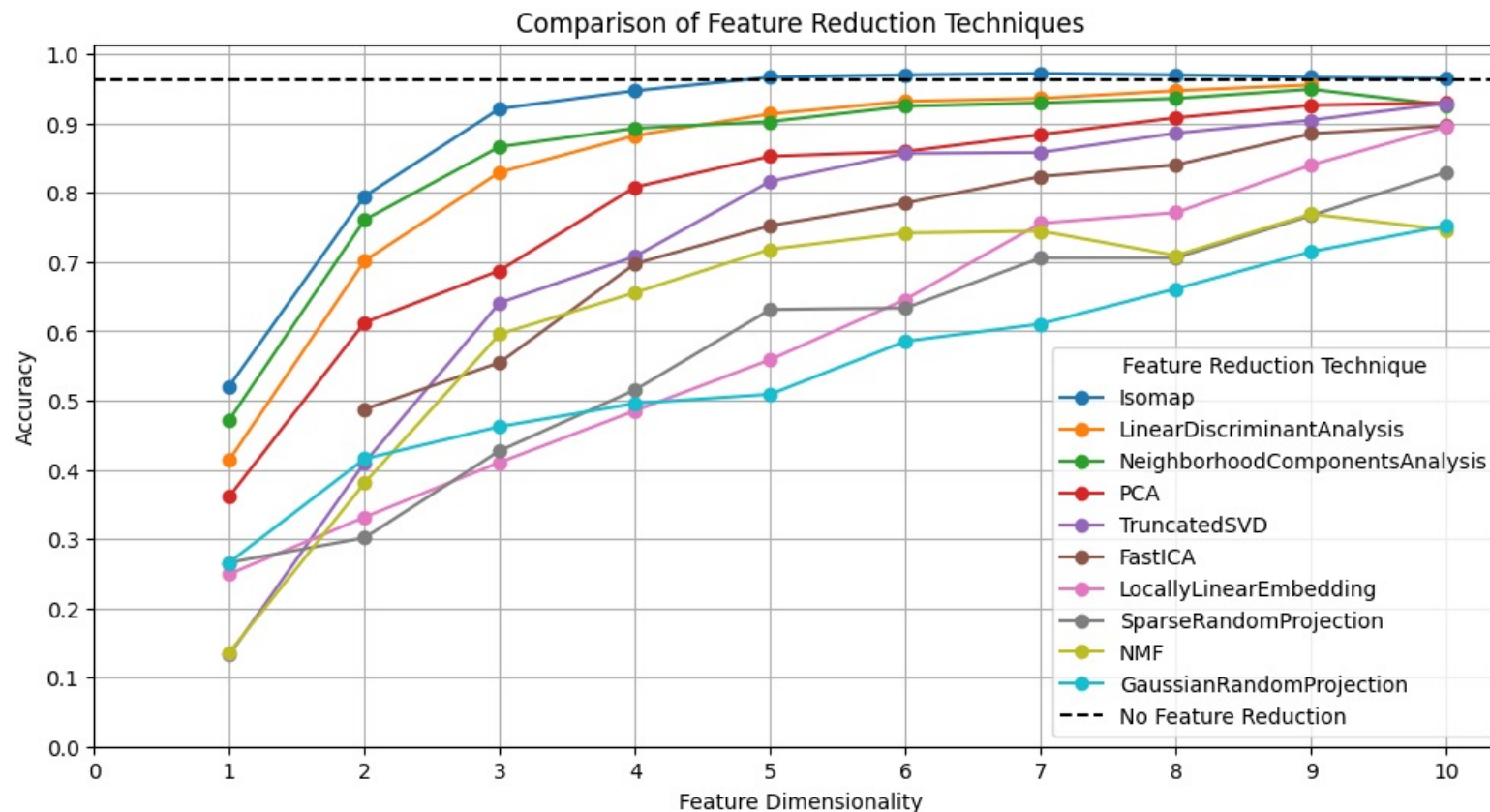
Since the classes are uniformly distributed, accuracy is an appropriate metric for model evaluation.

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

› FEATURE REDUCTION

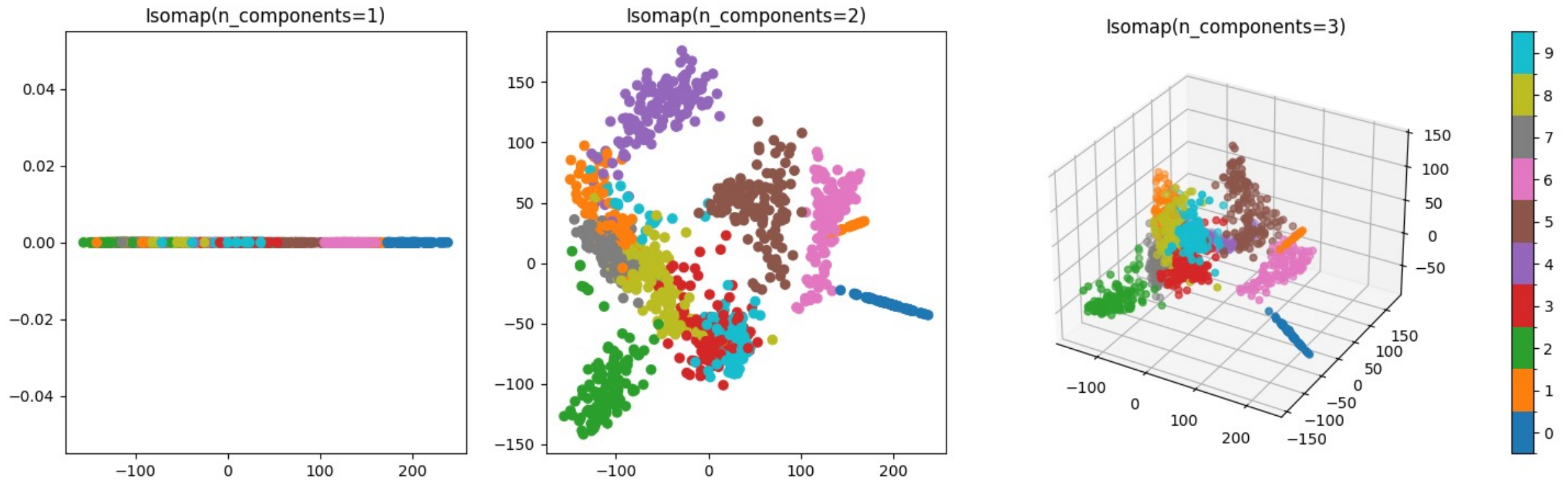
Trimming it down to the essentials.

FEATURE REDUCTION



Decision: Isomap seems to deliver the highest accuracy and starts converging at five dimensions.

FEATURE REDUCTION



› ENCODING CIRCUIT SELECTION

Getting data into the quantum state.

ENCODING CIRCUIT SELECTION

Optimization Problem

- > Maximize accuracy: Achieve the highest possible predictive performance
- > Minimize gate count: Reduce circuit depth to mitigate noise on real hardware

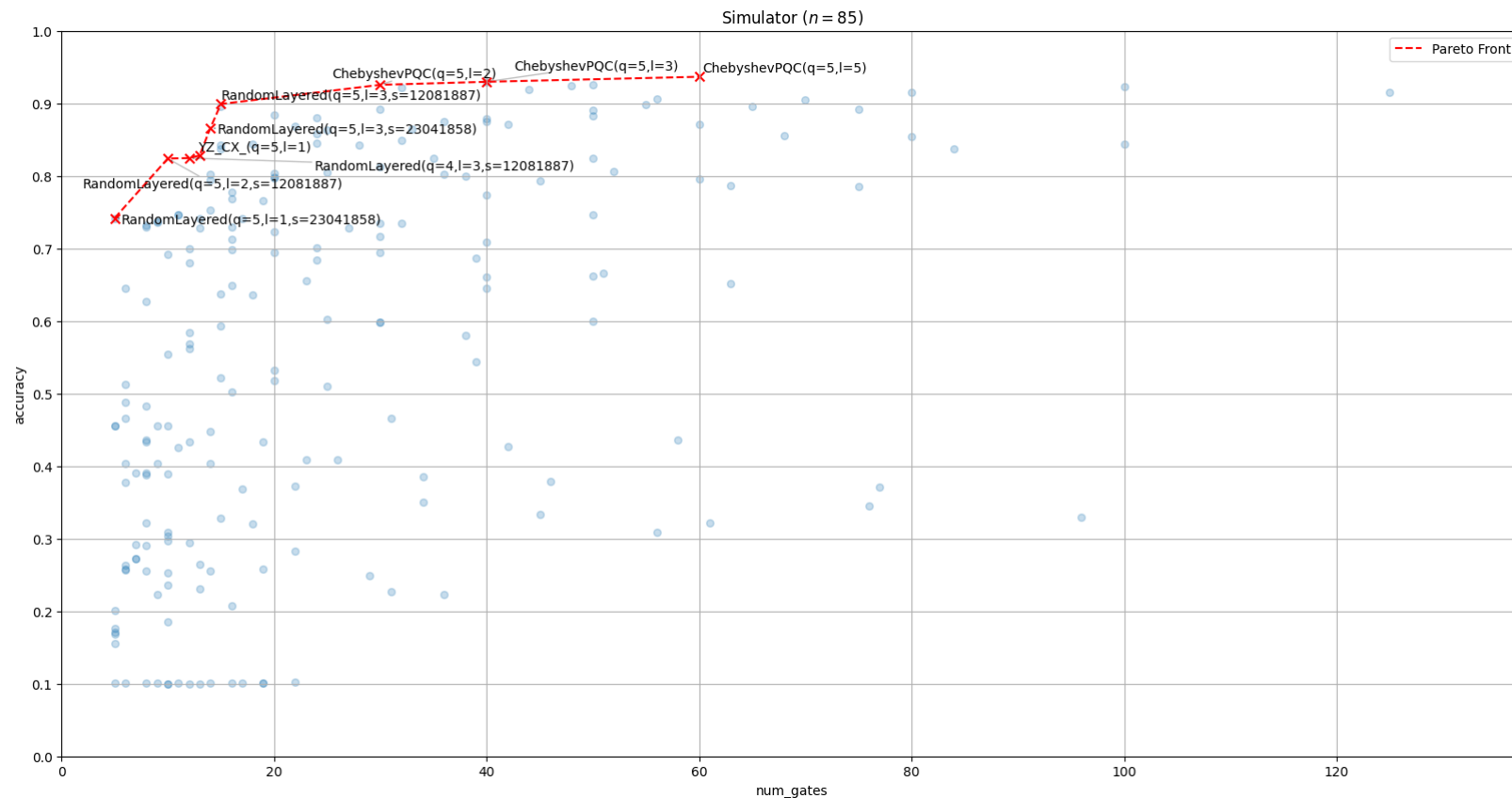
Types of Encoding Circuits

- > Predefined, parametrized circuits (e.g., `ChebyshevPQC`, `ParamZFeatureMap`, ...)
- > Randomly generated, non-parametrized circuits (e.g., `RandomLayeredEncodingCircuit`)

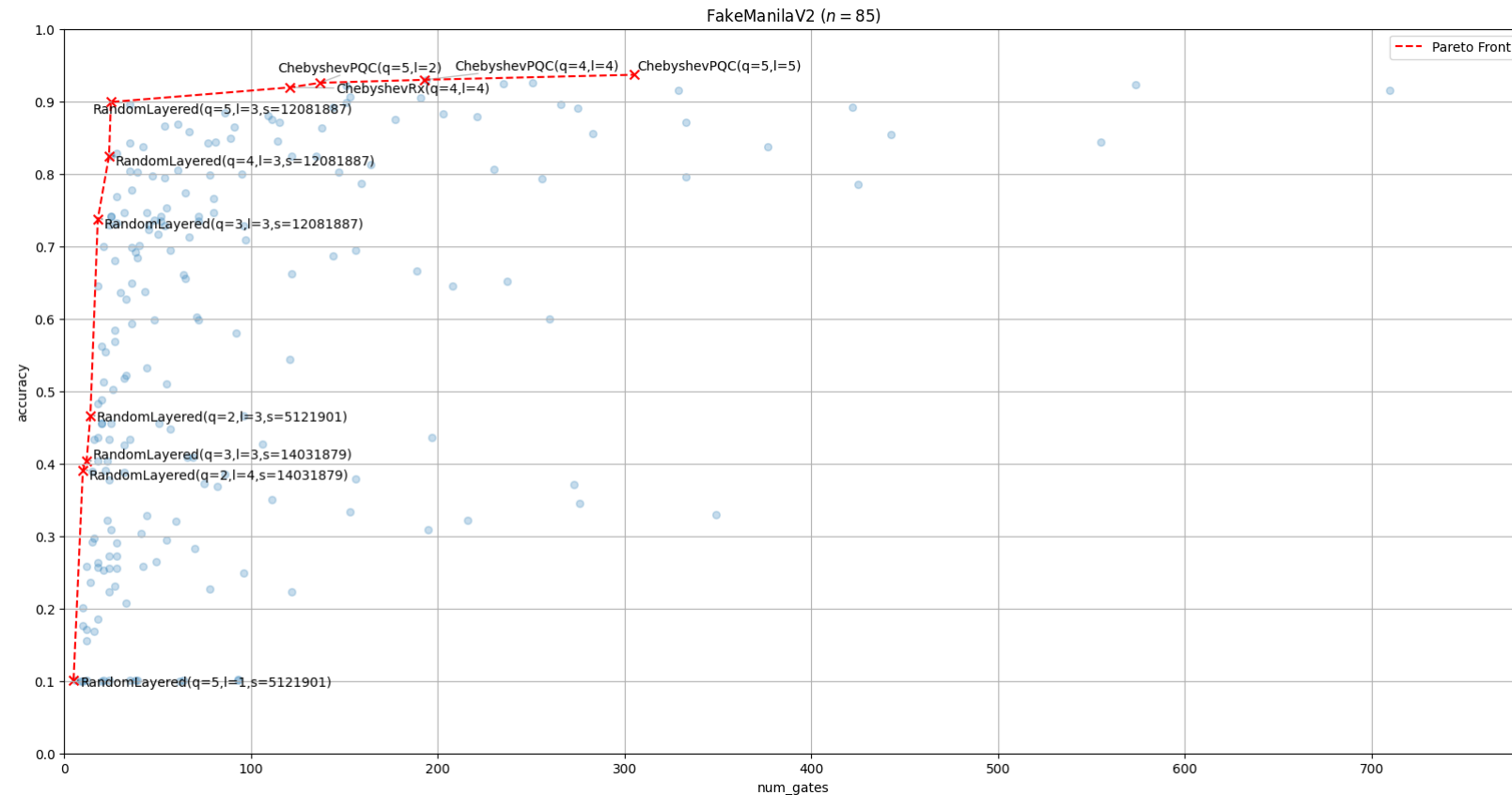
Hyperparameter Search

- > Each encoding circuit was tested with 1-5 qubits and 1-5 layers (25 combinations)
- > Random circuits were initialized using quantum physicists' birthdays as seeds

ENCODING CIRCUIT SELECTION



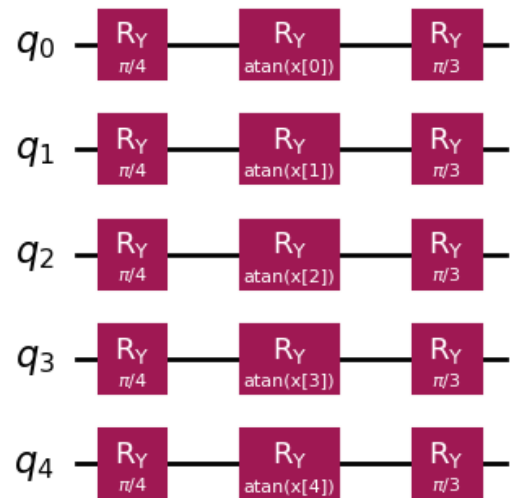
ENCODING CIRCUIT SELECTION



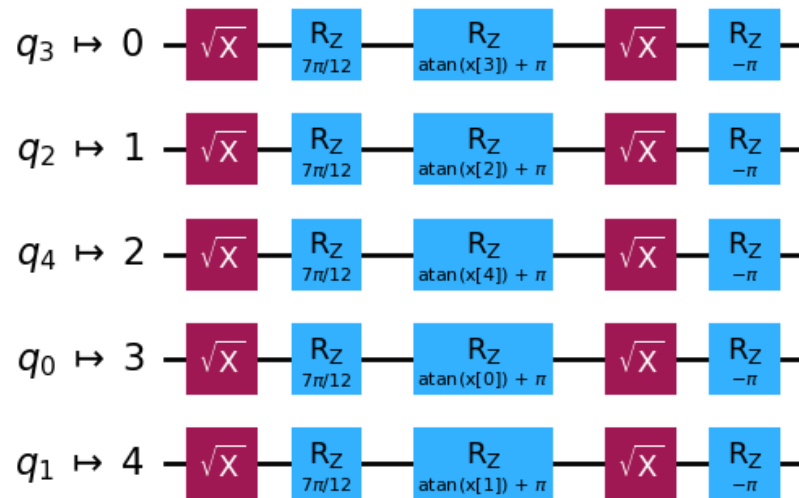
ENCODING CIRCUIT SELECTION

Decision: The random encoding circuit with 5 qubits, 3 layers, and seed 12081887 at the elbow of the Pareto front represents the optimal trade-off between accuracy and gate count.

Simulator



FakeManilaV2

Global Phase: $3\pi/2$ 

› OBSERVABLE SELECTION

Finding the best way to measure.

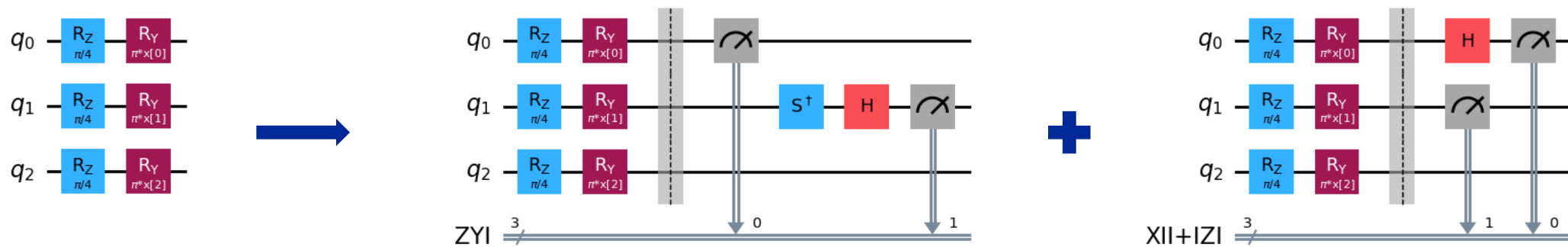
OBSERVABLE SELECTION

Optimization Problem

- > Maximize accuracy: Achieve the highest possible predictive performance
- > Minimize gate count: Reduce the complexity of the observable

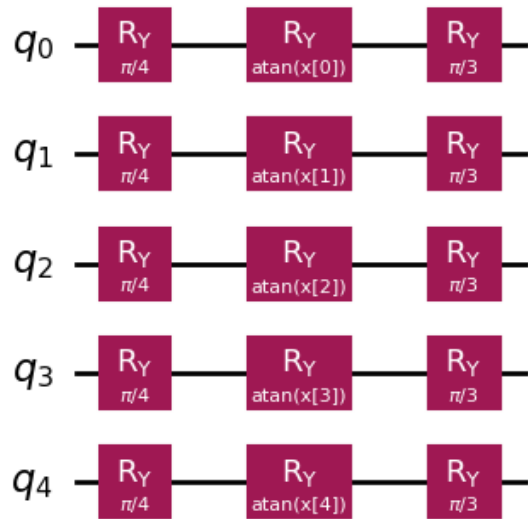
Example

- > $O = Z_1 Y_2 I_3 + X_1 I_2 I_3 + I_1 Z_2 I_3$

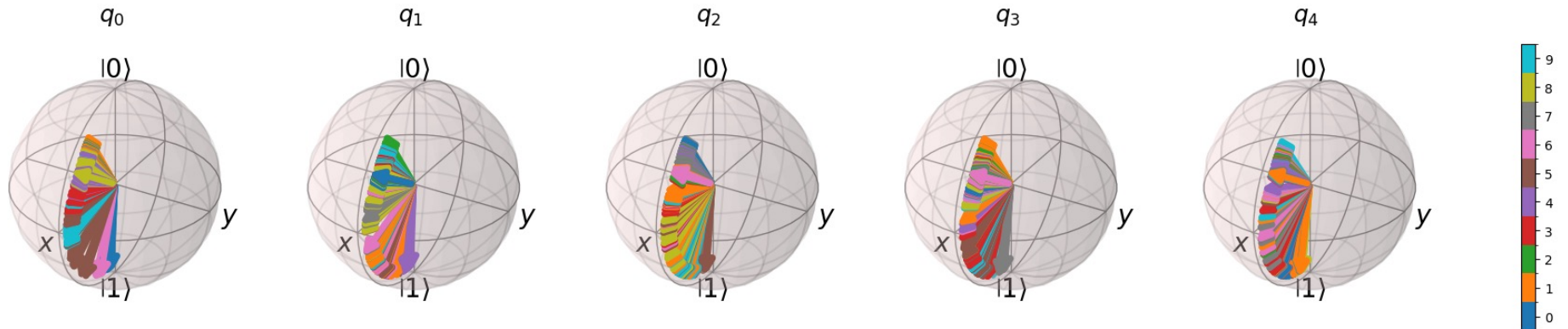


OBSERVABLE SELECTION

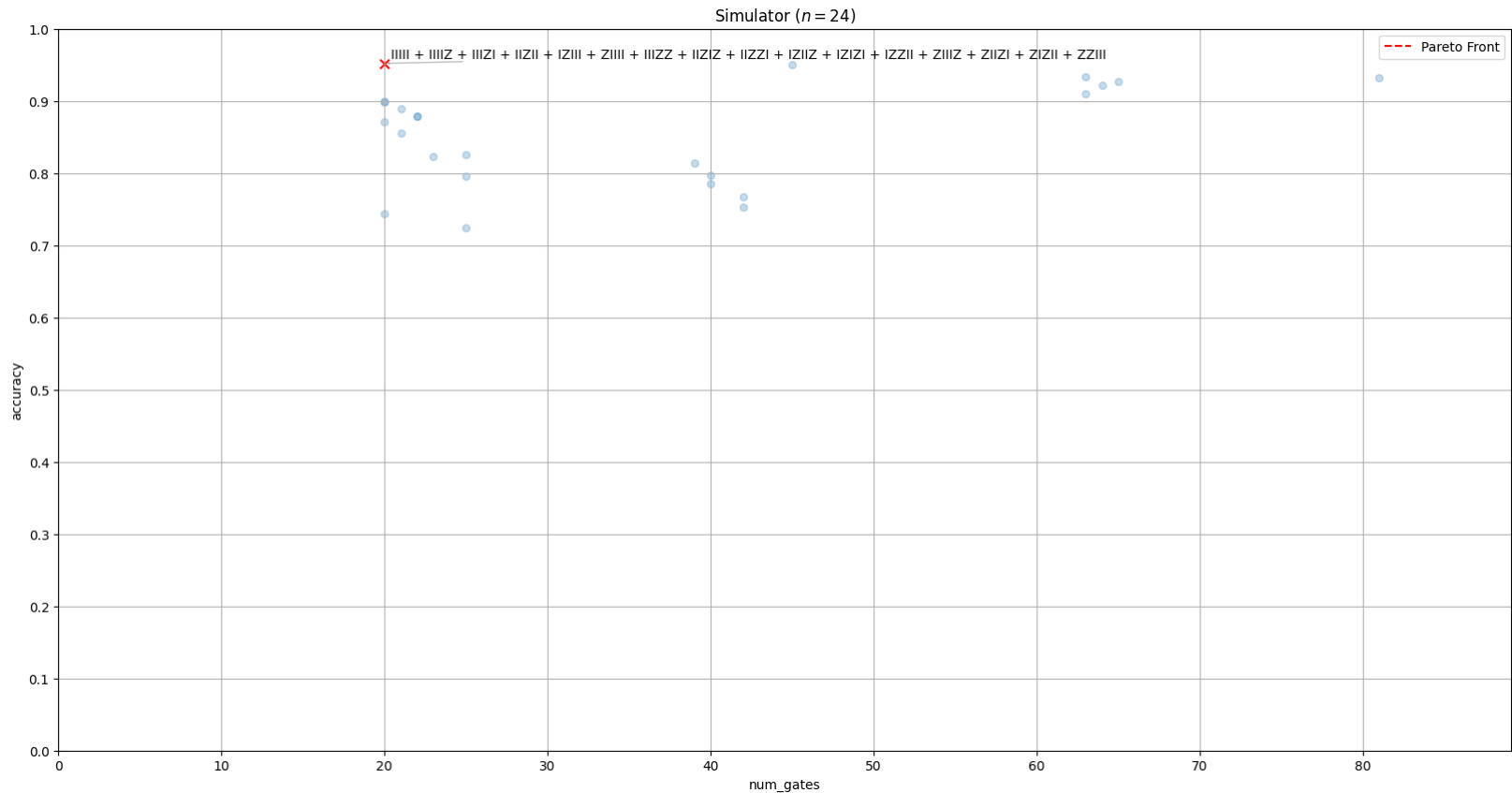
Preselection: The chosen encoding circuit consists only of R_Y gates, meaning the feature vector $\vec{x}$ is encoded only within the XZ plane of the Bloch sphere. Consequently, we can restrict the hyperparameter search to observables measuring in the X and Z bases.



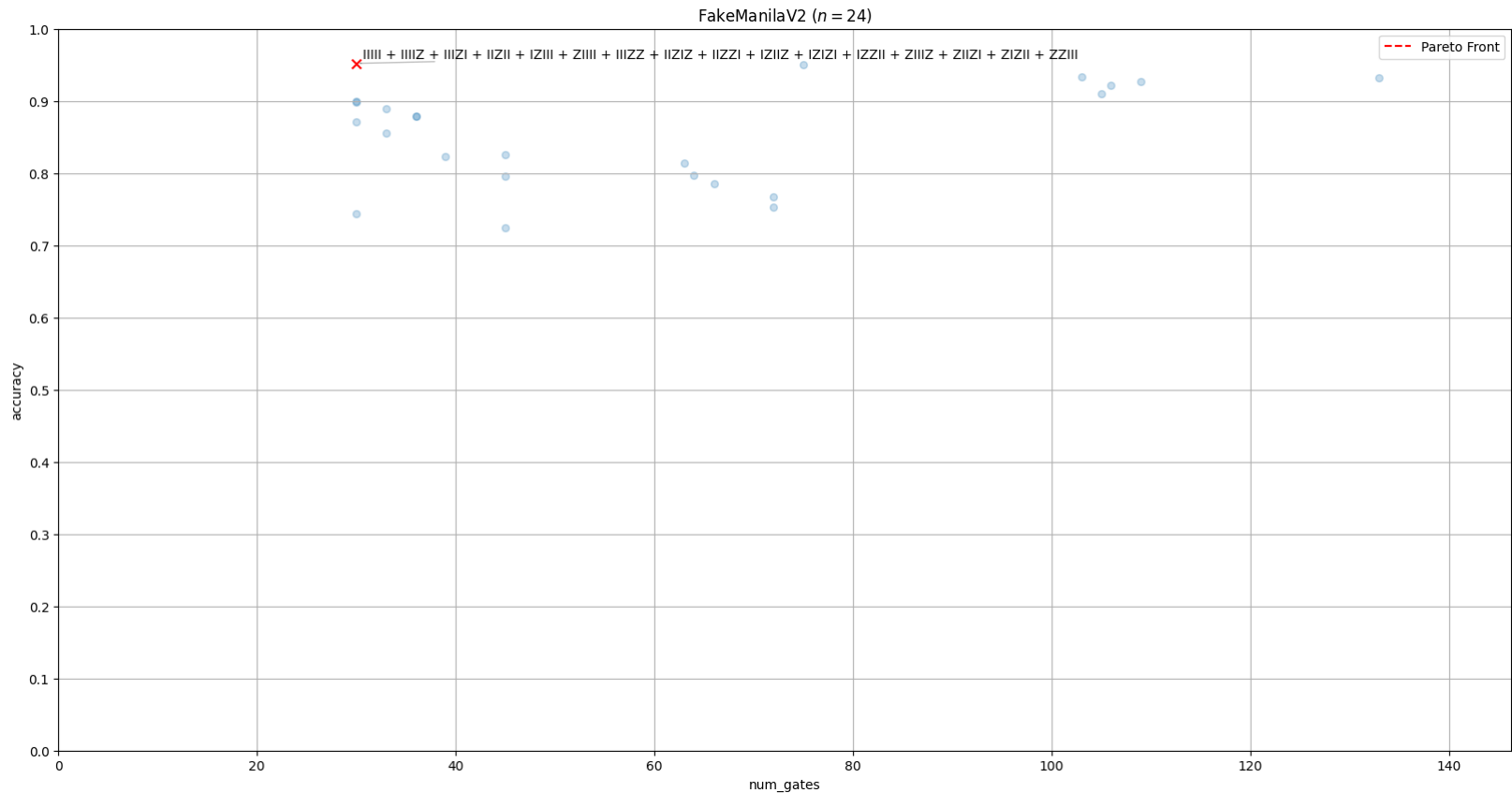
State vectors of the first 100 handwritten digits:



OBSERVABLE SELECTION

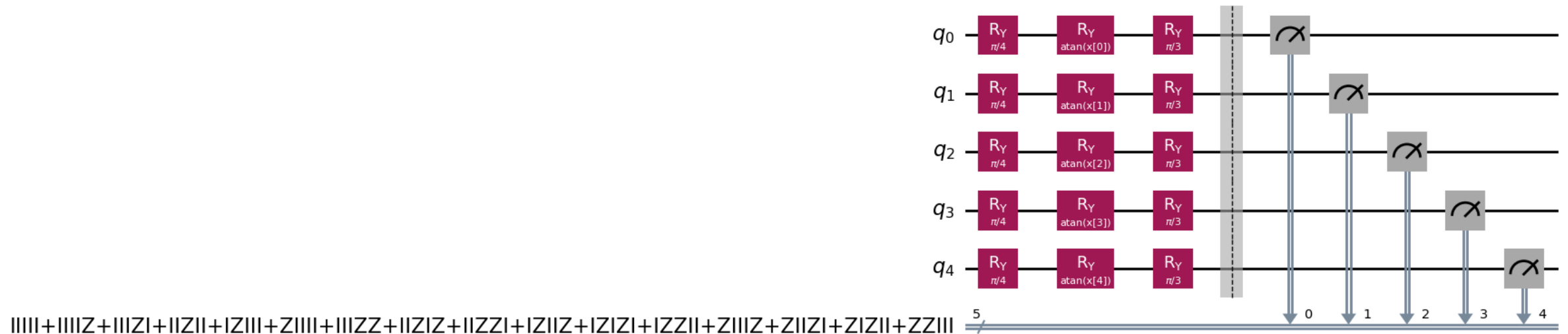


OBSERVABLE SELECTION



OBSERVABLE SELECTION

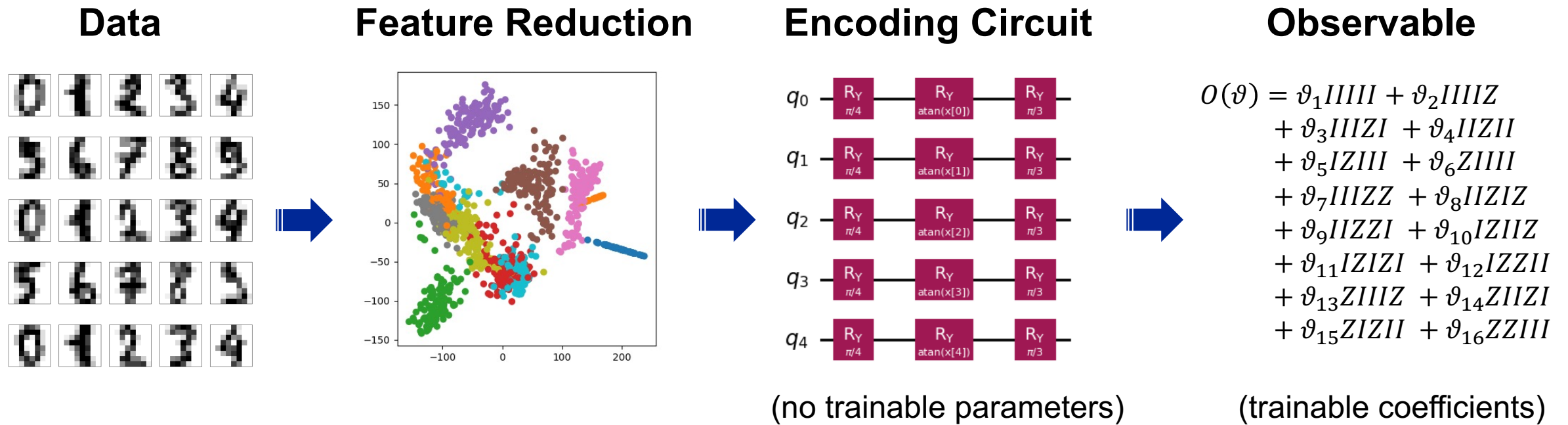
Decision: The Ising Hamiltonian is the optimal observable that measures every single qubit and all qubit pairs in the Z basis. Since all terms share the same basis, we can measure the entire observable using only a single circuit.



› FINAL PIPELINE & EVALUATION

Bringing it all together.

FINAL PIPELINE & EVALUATION



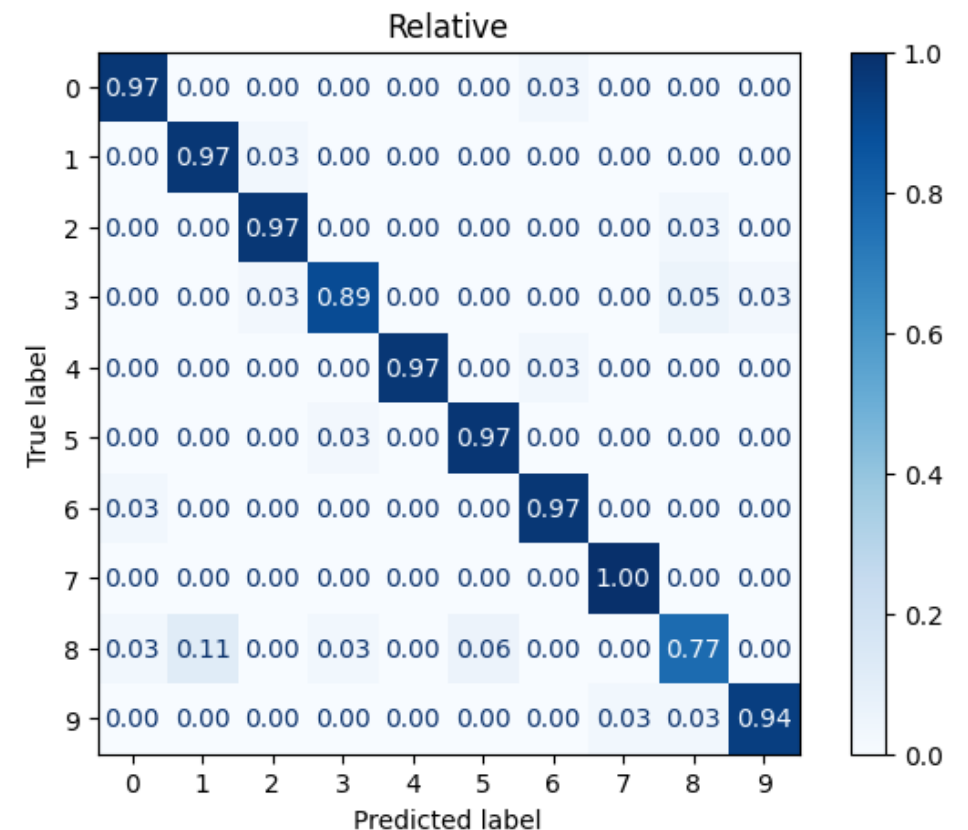
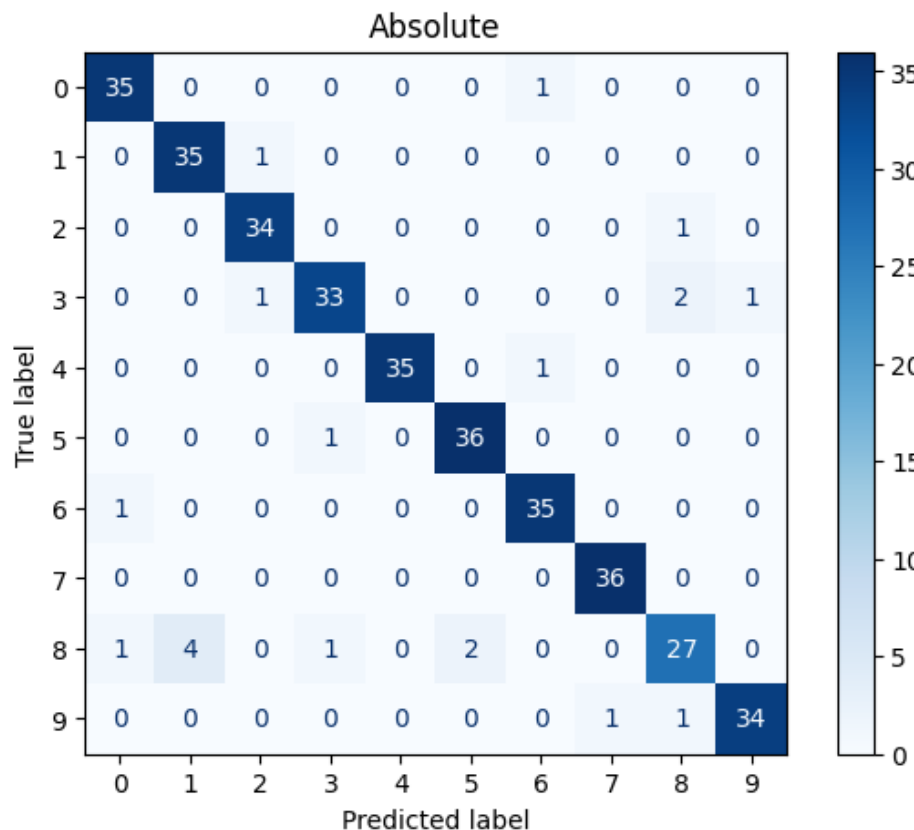
FINAL PIPELINE & EVALUATION

Accuracy

- > Train: 0.95
- > Test: 0.94

Parameters

- > Circuit: 0
- > Operator: 160



FINAL PIPELINE & EVALUATION

Criteria	Quantum Neural Network	Logistic Regression	
Train Accuracy	0.95	0.98	
Test Accuracy	0.94	0.97	
Feature Dimensionality	5	64	
HPO Duration	8 h + 4 h	1.5 s	
Training Duration	20 m	100 ms	
Parameters	160	 650	
Inference Latency	20 ms	20 ms	

> There is no quantum advantage for handwritten digits classification.

› (OPTIONAL) ACCELERATED HYPERPARAMETER SEARCH

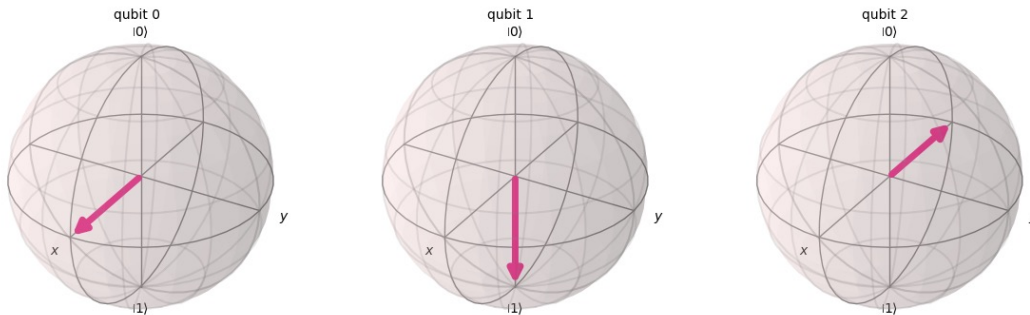
Speeding up the optimization.

(OPTIONAL) ACCELERATED HYPERPARAMETER SEARCH

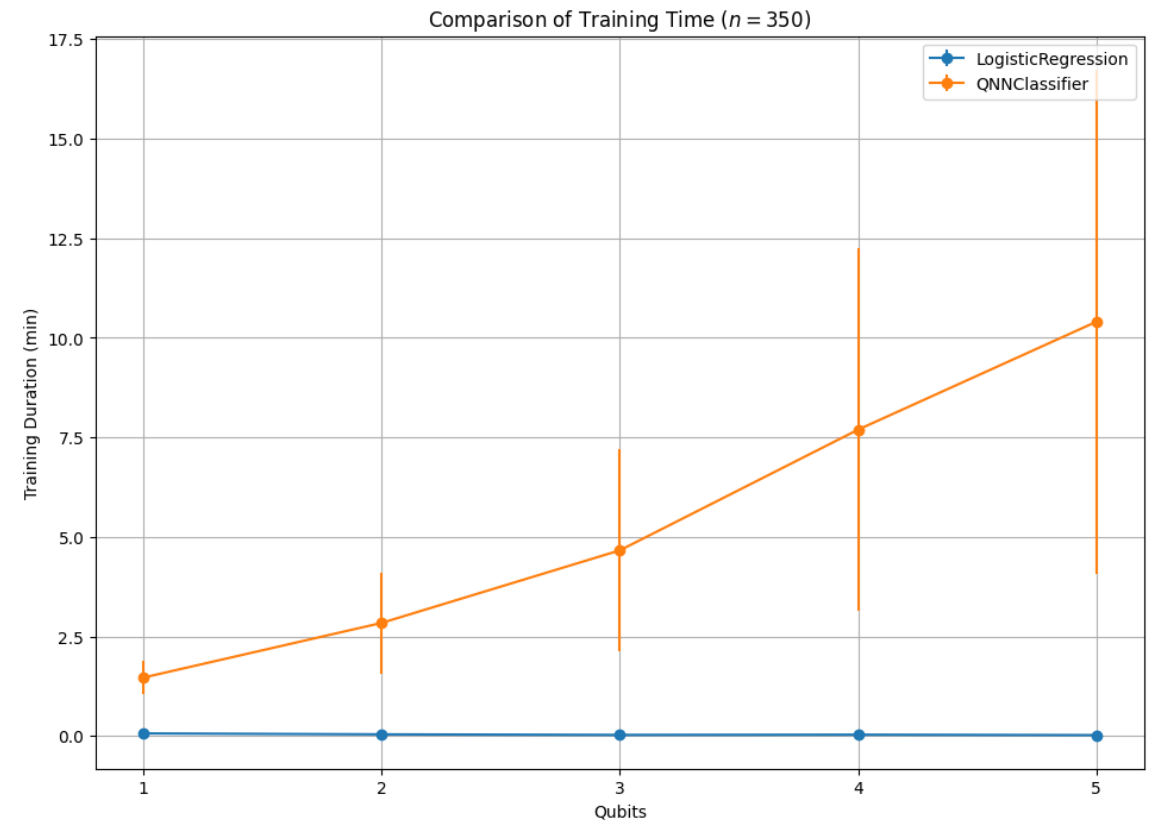
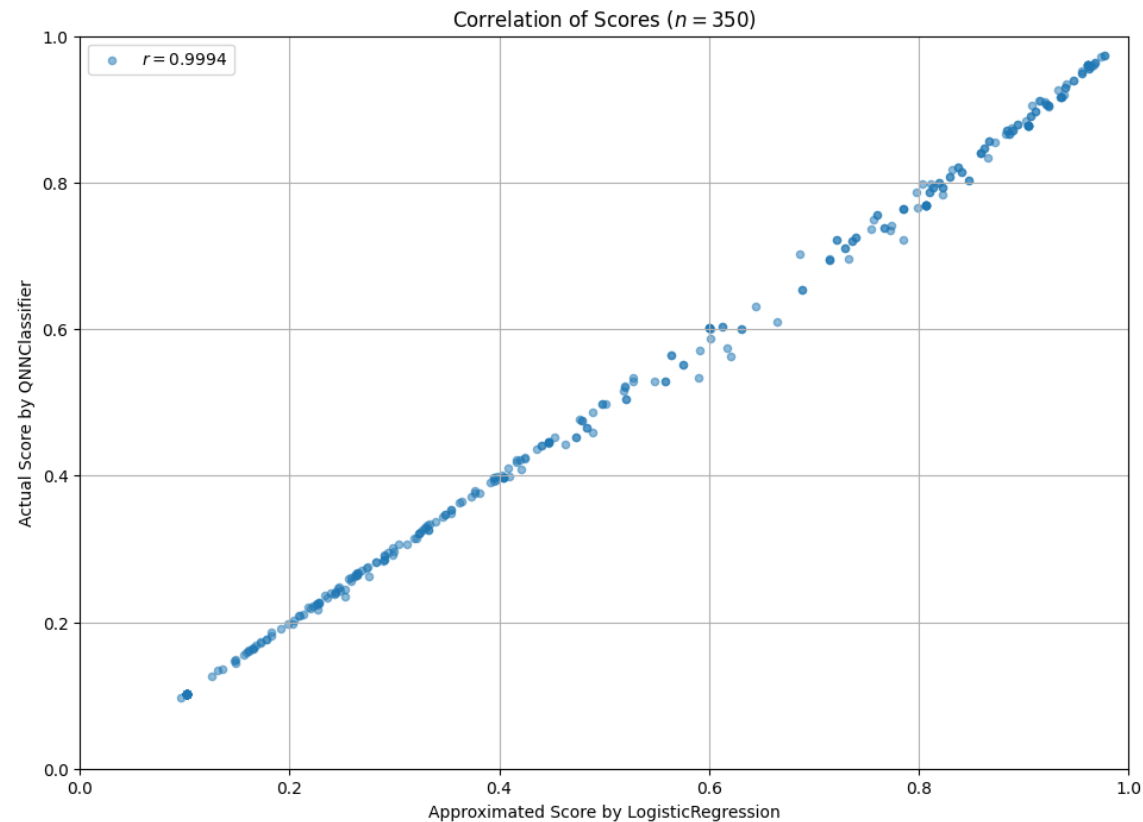
Hypothesis: We can approximately determine the accuracy of a non-parametrized encoding circuit and an observable by measuring the state vector and training a logistic regression model.

Example

- > Quantum: $c_1 ZII + c_2 IZI + c_3 IIZ + c_4 XII + c_5 IXI + c_6 IIX + c_7 III$
- > Classical: $w_1 z_1 + w_2 z_2 + w_3 z_3 + w_4 x_1 + w_5 x_2 + w_6 x_3 + b$



(OPTIONAL) ACCELERATED HYPERPARAMETER SEARCH



THANK YOU!

Please refer any questions to:



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